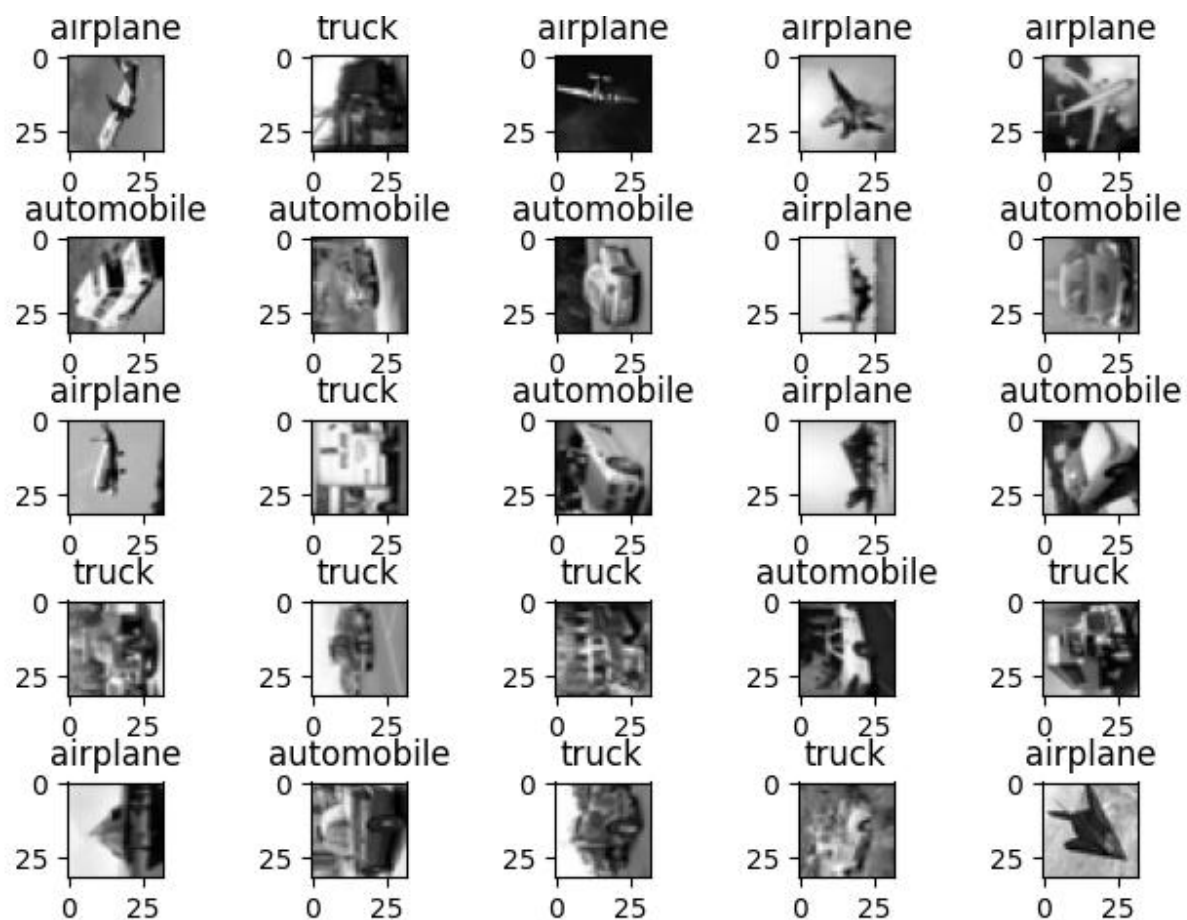


0: Preprocessing



ps6-0-a-1.png

1: Forward Propagation

1b) Accuracy: 0.7510666666666667

2: Cost Function

2b) Cost:

$\lambda = 0.1 : J = 1.1311919229773688$

$\lambda = 1 : J = 1.1474816945011665$

$\lambda = 2 : J = 1.1655814406387195$

3: Sigmoid Gradient

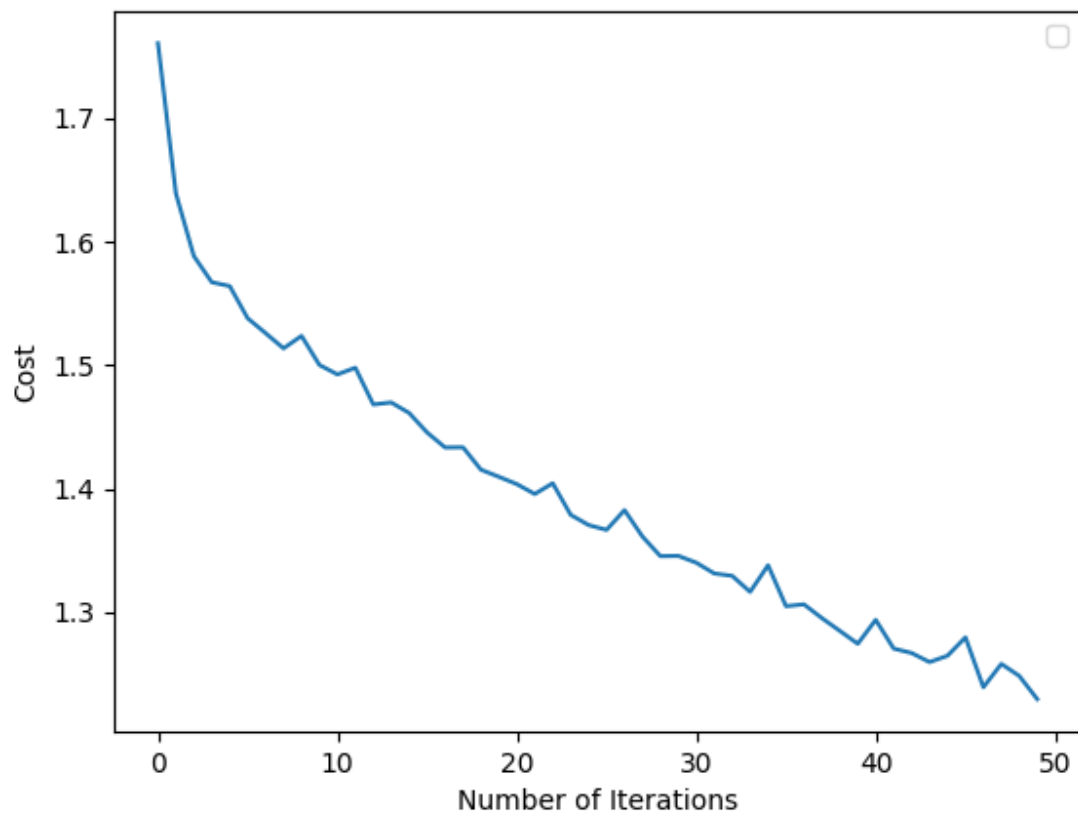
3) Gradient = $[[4.53958077e-05]$

[2.50000000e-01]

[4.53958077e-05]

4: Backpropagation

Alpha = 0.001



Ps6-4-e-1.png

5: Testing the Network

	MaxEpochs = 50		MaxEpochs = 300	
	Training Accuracy	Testing Accuracy	Training Accuracy	Testing Accuracy
Lambda = 0.01	0.71	0.67	0.93	0.66
Lambda = 0.1	0.71	0.68	0.92	0.66
Lambda = 0.2	0.71	0.66	0.92	0.66
Lambda = 1	0.70	0.68	0.91	0.67

	MaxEpochs = 50		MaxEpochs = 300	
	Training Cost	Testing Cost	Training Cost	Testing Cost
Lambda = 0.01	1.23	1.34	0.50	1.88
Lambda = 0.1	1.24	1.27	0.51	1.89
Lambda = 0.2	1.24	1.37	0.52	1.90
Lambda = 1	1.27	1.47	0.64	2.31

6: Discussion

The low performance of the testing accuracy seems to be because of overfitting, because the training accuracy gets to 90% with 300 epochs, but falls to 70% when applied to the testing data.